

Week 2

2.1 Coordinate Systems in $\mathbb{R}^2$ and $\mathbb{R}^3$

2.1.1 Polar Coordinates

Given a point $P = (x, y) \in \mathbb{R}^2$, we can represent it by the pair of values

$$\begin{cases} r = \sqrt{x^2 + y^2} = \text{the distance from the origin,} \\ \theta = \text{the anticlockwise angle between the x-axis and } \overrightarrow{OP}. \end{cases}$$

Example 25. The point $P = (1, 1)$ has $r = \sqrt{2}$ and $\theta = \frac{\pi}{4} + 2\pi k$, for $k \in \mathbb{Z}$.

We note that the value of θ is not uniquely defined. For the origin, $r = 0$, and θ may be any value.

Remark. There are different conventions for the ranges of r and θ . In this course we choose the convention $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$, so that (r, θ) are uniquely determined away from the origin. Although we make this convention, it is sometimes useful to allow $r < 0$ as we shall see later.

Given polar coordinates (r, θ) it is easy to read off the Cartesian coordinates directly

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Conversely, to read off the polar coordinates from the Cartesian coordinates, we have

$$r = \sqrt{x^2 + y^2}$$

For θ , there is not a unique representation, instead it varies by quadrant. For example, when $x > 0, y > 0$, then

$$\theta = \arctan\left(\frac{y}{x}\right).$$

There are equivalent expressions in the other four quadrants that ensure that $\theta \in [0, 2\pi)$.

2.1.2 Curves in Polar Coordinates

We explore some examples of curves in the plane represented using polar coordinates.

Example 26. The equation $r = r_0$ determines a circle of radius $r_0 > 0$ centred at the origin.

Example 27. The equation $\theta = \theta_0$ determines the half ray starting from (and including) the origin which makes an angle of θ_0 with the positive x -axis in the anticlockwise direction.

Example 28. The Archimedes spiral $r = k\theta$ for any constant $k > 0$.

Example 29. The equation $r = 4 \cos \theta$ determines a circle centred at $(2, 0)$ of radius 2. To see this we can change back to Cartesian coordinates. We first note that the origin satisfies this equation, so when multiplying by r we do not change the equation.

$$r = 4 \cos \theta \iff r^2 = 4r \cos \theta \iff x^2 + y^2 = 4x \iff (x - 2)^2 + y^2 = 2.$$

We note that under our convention that $r \geq 0$, we must restrict $\theta \in [0, \frac{\pi}{2}] \cup [\frac{3\pi}{2}, 2\pi)$.

Example 30. The equation $r \cos(\theta - \frac{\pi}{4}) = \sqrt{2}$ determines a straight line passing through $(0, -2)$ and $(2, 0)$. To see why we again change back to Cartesian coordinates. Expanding the right hand side using the double angle formula gives

$$2 = \sqrt{2}r \left(\cos \theta \cos \frac{\pi}{4} + \sin \theta \sin \frac{\pi}{4} \right) = r \cos \theta + r \sin \theta = x + y.$$

Recall that our convention is $r \geq 0$, however in some instances, allowing $r < 0$ can be convenient. In particular, for $r < 0$,

$$\begin{aligned} (x, y) &= (r \cos(\theta), r \sin(\theta)) \\ &= (-|r| \cos(\theta), -|r| \sin(\theta)) \\ &= (|r| \cos(\theta + \pi), |r| \sin(\theta + \pi)). \end{aligned}$$

Example 31. With this convention, the curve $\{\theta = \theta_0\}$ is now an entire straight line.

Example 32. Consider the curve $r = 1 - \lambda \cos \theta$ for some $\lambda > 1$.

Case 1: We begin with the convention $r \geq 0$.

$$1 - \lambda \cos \theta \geq 0 \implies \cos \theta \leq \lambda^{-1} < 1 \implies \theta \in [\delta, 2\pi - \delta],$$

for $\delta := \arccos(\lambda^{-1})$. Sketching this curve gives a simple cardioid curve.

Case 2: If we allow $r \in \mathbb{R}$ instead, then we have an extra loop for the region $\theta \in [-\delta, \delta]$. In particular, the curve is now closed and self-intersecting, i.e. not simple.

2.1.3 Coordinates in $\mathbb{R}^3$

One way to incorporate polar coordinates in higher dimensions is to use them within a plane, and leave the final Cartesian coordinate as it is. These are known as *cylindrical coordinates*.

To express a point $P = (x, y, z) \in \mathbb{R}^3$ in cylindrical coordinates, we convert the (x, y) part of P into polar coordinates. That is, we choose (r, θ, z) given by the relation

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

Example 33. The helix can be easily parameterised in cylindrical coordinates:

$$r = 1, \quad \theta = t, \quad z = t, \quad \forall t \in [0, 2\pi).$$

We also have a higher dimensional version of polar coordinates known as *spherical polar coordinates*. That is, we describe a point $P = (x, y, z) \in \mathbb{R}^3$ by:

$$\begin{cases} \rho = \sqrt{x^2 + y^2 + z^2} = \text{the distance from the origin,} \\ \theta = \text{the anticlockwise angle between the x-axis and } (x, y, 0), \\ \varphi = \text{the angle between the positive z-axis and } (x, y, z). \end{cases}$$

We note that $\varphi \in [0, \pi]$. To read off the spherical coordinates from the cartesian coordinates, we decompose $P = (x, y, 0) + (0, 0, z)$, and note that $z = \rho \cos \varphi$, and $(x, y, 0)$ has length $\rho \sin \varphi$. Therefore, we find that

$$x = \rho \sin \varphi \cos \theta, \quad y = \rho \sin \varphi \sin \theta, \quad z = \rho \cos \varphi.$$

Example 34. The subset $\rho = 2$ corresponds to the sphere centred at the origin of radius 2.

Example 35. The subset $\varphi = \frac{\pi}{4}$ corresponds to the cone with cone angle $\frac{\pi}{4}$, its tip at the origin, lying within the upper half space $\{z > 0\}$.

Example 36. The subset $\theta = 0$ corresponds to the half plane $\{x > 0, y = 0\}$.

2.2 Topological Terminology

2.2.1 Interior, Exterior and Boundary of a Set

Definition 2.8. Given a point $\vec{x}_0 \in \mathbb{R}^n$ and a fixed constant $r > 0$, we define the open ball centred at $\vec{x}_0$ of radius r to be

$$B_r(\vec{x}_0) := \{\vec{x} \in \mathbb{R}^n : \|\vec{x} - \vec{x}_0\| < r\}.$$

Similarly, we define the closed ball centred at $\vec{x}_0$ of radius r to be

$$\overline{B_r(\vec{x}_0)} := \{\vec{x} \in \mathbb{R}^n : \|\vec{x} - \vec{x}_0\| \leq r\}.$$

Definition 2.9. Let $S \subseteq \mathbb{R}^n$ be any subset.

- The interior of S is the set

$$\text{Int}(S) := \{\vec{x} \in \mathbb{R}^n : B_r(\vec{x}) \subseteq S, \text{ for some } r > 0\}.$$

Points in $\text{Int}(S)$ are called interior points of S .

- The exterior of S is the set

$$\text{Ext}(S) := \{\vec{x} \in \mathbb{R}^n : B_r(\vec{x}) \subseteq \mathbb{R}^n \setminus S, \text{ for some } r > 0\}.$$

Points in $\text{Ext}(S)$ are called exterior points of S .

- The boundary of S is the set

$$\partial S := \{\vec{x} \in \mathbb{R}^n : B_r(\vec{x}) \cap S \neq \emptyset \text{ and } B_r(\vec{x}) \cap (\mathbb{R}^n \setminus S) \neq \emptyset, \forall r > 0\}.$$

Points in ∂S are called boundary points of S .

Example 37. Consider the annulus $S = \{(r, \theta) \in \mathbb{R}^2 : 1 < r \leq 4\}$, where (r, θ) denotes polar coordinates.

Let A be a point lying on the inner circle $r = 1$, B be a point lying inside the annulus with $r \in (1, 4)$, C be a point on the outer circle $r = 4$, and D a point with $r > 4$.

We note that $A, D \notin S$, $B, C \in S$, and that

- $A \in \partial S$ is a boundary point of S ;
- $B \in \text{Int}(S)$ is an interior point of S ;
- $C \in \partial S$ is a boundary point of S ;
- $D \in \text{Ext}(S)$ is an exterior point of S .

Let $S \subseteq \mathbb{R}^n$. The following properties follow directly from the definition:

1. $\mathbb{R}^n$ is the disjoint union of $\text{Int}(S)$, $\text{Ext}(S)$ and ∂S ;
2. $\text{Int}(S) \subseteq S$ and $\text{Ext}(S) \subseteq \mathbb{R}^n \setminus S$. A point in ∂S may or may not be in S .

2.2.2 Open and Closed Sets

Definition 2.10. A subset $S \subseteq \mathbb{R}^n$ is called

1. open if $\forall \vec{x} \in S, \exists \varepsilon > 0$ such that $B_\varepsilon(\vec{x}) \subseteq S$.
2. closed if $\mathbb{R}^n \setminus S$ is open.

Equivalently, S is open if $S = \text{Int}(S)$, and S is closed if $S = \text{Int}(S) \cup \partial S$.

Example 38. For $\vec{y} \in \mathbb{R}^n$, the open ball $S := B_r(\vec{y}) \subset \mathbb{R}^n$ is open (as the name suggests).

To prove this, we choose a point $\vec{x} \in S$ and show there exists $\varepsilon > 0$ such that $B_\varepsilon(\vec{x}) \subset S$.

Since $\vec{x} \in B_r(\vec{y})$, we know that

$$\|\vec{x} - \vec{y}\| < r,$$

let's say that $\|\vec{x} - \vec{y}\| = a$. Furthermore, if $\varepsilon > 0$ and $\vec{z} \in B_\varepsilon(\vec{x})$, then

$$\|\vec{x} - \vec{z}\| < \varepsilon.$$

Therefore, by the triangle inequality:

$$\|\vec{z} - \vec{y}\| \leq \|\vec{z} - \vec{x}\| + \|\vec{x} - \vec{y}\| < a + \varepsilon.$$

If we choose $\varepsilon = r - a$, it then follows that $\|\vec{z} - \vec{y}\| < r$, and so we have $\vec{z} \in B_r(\vec{y}) = S$. We have therefore shown that $B_\varepsilon(\vec{x}) \subset S$, which is what it means for S to be open.

Example 39. By a similar argument, we may show that the closed ball $\overline{B_r(\vec{y})}$ is closed.

Remark. There are precisely two subsets of $\mathbb{R}^n$ which are both open and closed: $\mathbb{R}^n$ and $\emptyset$. Some subsets of $\mathbb{R}^n$ are neither open nor closed, e.g the annulus $\{1 < r \leq 4\}$ from before.

Note that, for any $S \subseteq \mathbb{R}^n$, then it is always true that $\text{Int}(S)$ and $\text{Ext}(S)$ are both open, and hence ∂S is closed.

2.2.3 Bounded and Path Connected Subsets

Definition 2.11. $S \subseteq \mathbb{R}^n$ is bounded if $\exists M > 0$ such that

$$S \subseteq B_M(0) = \{\vec{x} \in \mathbb{R}^n : \|\vec{x}\| < M\}.$$

S is unbounded if S is not bounded.

Definition 2.12. S is path connected if any two points in S can be connected by a curve in S , i.e. $\forall \vec{x}, \vec{y} \in S$, there exists a curve $\gamma : [0, 1] \rightarrow S$ with $\gamma(0) = \vec{x}$ and $\gamma(1) = \vec{y}$.

Remark. There is also a notion of connectedness which is slightly more subtle that we shall not discuss in these notes. It is a fact that if S is path connected, then S is connected, however the converse does not hold: there exists sets S which are connected but not path connected.

Theorem 40 (Jordan curve theorem). A simple closed curve in $\mathbb{R}^2$ divides $\mathbb{R}^2$ into two path connected components, one bounded and one unbounded.

Remark. Although this seems trivial, the result is surprisingly hard to prove. In fact, an analogous statement in higher dimensions which was believed to be true for many years is actually false. (For the interested reader, you may look up the Jordan–Schönflies theorem and the Alexander horned sphere.)

2.3 Vector-Valued Multivariable Functions

We now consider functions $\vec{f} : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$. One way to visualise such a function is by looking at the graph of $\vec{f}$:

$$\text{Graph}(\vec{f}) = \{(\vec{x}, \vec{f}(\vec{x})) \in \mathbb{R}^{n+m} : \vec{x} \in \Omega\}.$$

Alternatively, we may consider the level sets of $\vec{f}$: for any $\vec{c} \in \mathbb{R}^m$, we define the level set at $\vec{c}$ to be

$$L_{\vec{c}} := \vec{f}^{-1}(\vec{c}) = \{x \in \Omega : \vec{f}(x) = \vec{c}\} \subseteq \Omega \subseteq \mathbb{R}^n.$$

Example 41. For the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x + y$, and for any $c \in \mathbb{R}$, we have that the level set L_c is the line given by the equation $y = c - x$.

Example 42. For the function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$, $g(x, y) = x^2 + y^2$, and for any $c \in \mathbb{R}$, we have that

$$L_c = \begin{cases} \emptyset & : c < 0 \\ (0, 0) & : c = 0 \\ \text{the circle } \partial B_{\sqrt{c}}(0) & : c > 0. \end{cases}$$

Example 43. For the function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$, $h(x, y) = \cos(2\pi(x^2 + y^2))$, the level set L_1 is given by those points (x, y) such that $x^2 + y^2$ is an integer. Therefore

$$L_1 = \{0\} \cup \bigcup_{k \geq 1} \partial B_{\sqrt{k}}(0).$$

2.3.1 Limits of Multivariable Functions

Let $A \subseteq \mathbb{R}^n$. We define $\bar{A} = A \cup \partial A$, known as the *closure* of A . Since $A \subseteq \text{Int}(A) \cup \partial A$, we see that

$$\bar{A} = \text{Int}(A) \cup \partial A = \mathbb{R}^n \setminus \text{Ext}(A),$$

and hence $\bar{A}$ is closed.

Fact: $\bar{A}$ is the smallest closed set containing A .

For any point $\vec{a} \in \bar{A}$ and any function $\vec{f} : A \rightarrow \mathbb{R}^m$, we now define a notion of a limit of $\vec{f}(\vec{x})$ as $\vec{x} \rightarrow \vec{a}$.

Definition 2.13. For $\vec{f} : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\vec{a} \in \bar{A}$ and $\vec{L} \in \mathbb{R}^m$, we say that $\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) = \vec{L}$ if, $\forall \epsilon > 0, \exists \delta > 0$ such that,

$$\vec{x} \in A \quad \text{with} \quad 0 < \|\vec{x} - \vec{a}\| < \delta \quad \implies \quad \|\vec{f}(\vec{x}) - \vec{L}\| < \epsilon.$$

Remark. This is the same definition of a limit as in the 1-dimensional case, only with the absolute value $|\cdot|$ being replaced by the length of vectors $\|\cdot\|$. Here, you should read the quantity $\|\vec{x} - \vec{y}\|$ as the distance between $\vec{x}$ and $\vec{y}$.

The fact that we require $\|\vec{x} - \vec{a}\| > 0$ means we do not care about the value of $\vec{f}(\vec{a})$ if it exists.

Example 44. Take $f(x, y) = x + y$. We will prove using the definition that

$$\lim_{(x,y) \rightarrow (2,1)} f(x, y) = 3.$$

That is, we must show that for any $\epsilon > 0$, we can find a $\delta > 0$ (which may depend on ϵ) such that, if

$$0 < \|(x, y) - (2, 1)\| < \delta,$$

then

$$|f(x, y) - 3| = |x + y - 3| < \epsilon.$$

We first notice that the quantity we want to control can be bound above using the triangle inequality:

$$|x + y - 3| \leq |x - 2| + |y - 1|. \quad (2.8)$$

Moreover, we have that each of these terms is controlled by the distance between (x, y) and $(2, 1)$:

$$|x - 2|, |y - 1| \leq \sqrt{(x - 2)^2 + (y - 1)^2} = \|(x, y) - (2, 1)\|. \quad (2.9)$$

Therefore, if we picked $\epsilon = 1$ for example, then choosing $\delta = 1/2$ we see that

$$0 < \|(x, y) - (2, 1)\| < 1/2$$

implies by (2.9) that

$$|x - 2|, |y - 1| < 1/2 \implies |x - 2| + |y - 1| < 1,$$

which by (2.8) that

$$|x + y - 3| < 1,$$

as required.

More generally, if we fix $\epsilon > 0$ to be an arbitrary positive number, we then choose $\delta = \epsilon/2 > 0$. It then follows from (2.8) and (2.9) that if $0 < \|(x, y) - (2, 1)\| < \delta$, then $|x + y - 3| < \epsilon$. Since the initial $\epsilon > 0$ chosen was arbitrarily, we have shown it to be true for any $\epsilon > 0$.

Example 45. Take $f(x, y) = x^2 + y^2$. We will prove using the definition that

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0.$$

Fix $\epsilon > 0$ and choose $\delta = \sqrt{\epsilon} > 0$. If $0 < \|(x, y) - (0, 0)\| = \|(x, y)\| < \delta$, we have

$$|f(x, y) - 0| = |x^2 + y^2| = \|(x, y)\|^2 < \delta^2 = \epsilon.$$

2.3.2 Component Functions

Let $A \subseteq \mathbb{R}^n$, $\vec{a} \in \bar{A}$, $\vec{f} : A \rightarrow \mathbb{R}^m$. We can decompose the vector $\vec{f}(\vec{x})$ at every point $\vec{x} \in A$:

$$\vec{f}(\vec{x}) = \begin{pmatrix} f_1(\vec{x}) \\ f_2(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{pmatrix},$$

where each $f_i : A \rightarrow \mathbb{R}$ for $i = 1, \dots, m$ is called a *component* of $\vec{f}$.

Lemma 46. Let $\vec{f} : A \rightarrow \mathbb{R}^m$, $\vec{a} \in \bar{A}$ and $\vec{L} \in \mathbb{R}^m$. Then

$$\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) = \vec{L} = (\ell_1, \dots, \ell_m) \in \mathbb{R}^m \iff \lim_{\vec{x} \rightarrow \vec{a}} f_j(\vec{x}) = \ell_j, \quad \forall j = 1, \dots, m.$$

As a consequence of this lemma, it is enough to focus on limits of $\mathbb{R}$ -valued functions ($m = 1$).

Proof. We begin with the ($\implies$) direction. Fix $\epsilon > 0$. By the definition of $\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) = \vec{L}$, there exists $\delta > 0$ such that $\vec{x} \in A$ with $0 < \|\vec{x} - \vec{a}\| < \delta$ implies that

$$|f_j(\vec{x}) - \ell_j| \leq \sqrt{\sum_{i=1}^k (f_i(\vec{x}) - \ell_i)^2} = \|\vec{f}(\vec{x}) - \vec{L}\| < \epsilon,$$

for each $j = 1, \dots, m$. This is precisely what we wanted to show.

For the reverse direction ($\impliedby$), we again fix $\epsilon > 0$. Consider the new value $\tilde{\epsilon} = \frac{\epsilon}{\sqrt{m}} > 0$. For each j , plugging this value of $\tilde{\epsilon}$ into the definition of $\lim_{\vec{x} \rightarrow \vec{a}} f_j(\vec{x}) = \ell_j$, we have that there exists $\delta_j > 0$ such that, if $\vec{x} \in A$ and if $0 < \|\vec{x} - \vec{a}\| < \delta_j$, then $|f_j(\vec{x}) - \ell_j| < \tilde{\epsilon}$.

We now set $\delta := \min\{\delta_1, \dots, \delta_m\} > 0$. Then, if $\vec{x} \in A$ and if $0 < \|\vec{x} - \vec{a}\| < \delta$, we have that

$$\|\vec{f}(\vec{x}) - \vec{L}\| = \sqrt{\sum_{j=1}^m |f_j(\vec{x}) - \ell_j|^2} \leq \sqrt{\sum_{j=1}^m \tilde{\epsilon}^2} = \sqrt{m} \tilde{\epsilon} = \epsilon.$$

This is precisely the definition of $\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) = \vec{L}$.

□

2.3.3 Paths in the Domain

When the domain is a single variable, the ways we may approach at point $a \in \mathbb{R}$ are limited. In particular, there are only two directions to consider; from above and from below. Therefore

$$\lim_{x \rightarrow a} f(x) \text{ exists} \iff \underbrace{\lim_{x \uparrow a} f(x) = \lim_{x \downarrow a} f(x)}_{\text{(they exist and are equal)}}.$$

When our domain is in $\mathbb{R}^n$ for $n \geq 2$, there are now infinitely many ways to approach a point $\vec{a} \in \mathbb{R}^n$. When defining a limit, we need to consider *every* possible curve along which we can approach $\vec{a}$. In particular, $\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) = \vec{L}$ if and only if, the limit of $\vec{f}(\vec{x})$ when $\vec{x}$ approaches $\vec{a}$ along any path exists and equals $\vec{L}$.

Therefore, in order to show that such a limit does *not* exist, we can either:

- Find a path along which the limit does not exist;
- Find two paths such that the limit along these paths is different.

In both cases, we may conclude that the limit does not exist (DNE).

Example 47. Consider the function $f : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$, $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$. In order to investigate the limit of f at the origin, we look at the limits of f along different paths to the origin.

(i) Along the x -axis, we have

$$\lim_{(x,0) \rightarrow (0,0)} \frac{x^2 - 0}{x^2 + 0} = \lim_{x \rightarrow 0} 1 = 1.$$

(ii) Along the y -axis, we have

$$\lim_{(0,y) \rightarrow (0,0)} \frac{0 - y^2}{0 + y^2} = \lim_{y \rightarrow 0} -1 = -1.$$

Since $1 \neq -1$, we conclude that the limit $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

Example 48. Consider the function $g : \mathbb{R}^2 \setminus \{(1, 2)\} \rightarrow \mathbb{R}$, $g(x, y) = \frac{xy - 2x - y + 2}{(x-1)^2 + (y-2)^2}$. To investigate the limit of g at $(1, 2)$, we approach $(1, 2)$ along the line with slope $m \in \mathbb{R}$. i.e along

$$y = 2 + m(x - 1).$$

We first note that

$$g(x, y) = \frac{xy - 2x - y + 2}{(x-1)^2 + (y-2)^2} = \frac{(x-1)(y-2)}{(x-1)^2 + (y-2)^2},$$

and so

$$g(x, m(x-1) + 2) = \frac{m(x-1)^2}{(x-1)^2 + m^2(x-1)^2}.$$

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Therefore, taking a limit along this line gives

$$\begin{aligned}\lim_{(x, m(x-1)+2) \rightarrow (1,2)} g(x, m(x-1)+2) &= \lim_{x \rightarrow 1} \frac{m(x-1)^2}{(x-1)^2 + m^2(x-1)^2} \\ &= \lim_{x \rightarrow 1} \frac{m}{1+m^2} = \frac{m}{1+m^2}.\end{aligned}$$

Since this value is different for different choices of $m \in \mathbb{R}$, the limit $\lim_{(x,y) \rightarrow (1,2)} g(x,y)$ DNE.

Example 49. $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined by

$$f(x, y) = \begin{cases} 1 & : 0 < y < x^2 \\ 0 & : \text{else.} \end{cases}$$

Investigate the limit of f at the points

$$\vec{a} = (0, 1), \quad \vec{b} = (1, 1), \quad \vec{c} = (0, 0).$$

- Near the point $\vec{a}$, $f \equiv 0$, and so

$$\lim_{(x,y) \rightarrow \vec{a}} f(x, y) = 0.$$

- We note that

$$f(x, 1) = \begin{cases} 1 & : x > 1 \\ 0 & : x \leq 1. \end{cases}$$

Therefore,

$$\lim_{x \downarrow 1} f(x, 1) = 0, \quad \lim_{x \uparrow 1} f(x, 1) = 1,$$

and since these are different the limit of f at $\vec{b}$ DNE.

- Along any line $y = mx$ with $m > 0$ we have that

$$f(x, mx) = \begin{cases} 1 & : x > m \\ 0 & : x \leq m. \end{cases}$$

Therefore, $\lim_{(x,mx) \rightarrow (0,0)} f(x, mx) = 0$, for all $m > 0$. A similar calculation shows the same for $m \leq 0$.

However, this does **not** mean that the limit at $\vec{c}$ exists and is zero. To see why, consider instead the curve $y = \frac{1}{2}x^2$. At every point on this curve (except the origin) $f \equiv 1$. Therefore,

$$\lim_{(x, \frac{x^2}{2}) \rightarrow (0,0)} f(x, \frac{x^2}{2}) = 1 \neq 0,$$

and so the limit of f at $\vec{c}$ DNE.

This example demonstrates that it is not enough to simply check the straight line paths to the limit point to verify the value of a limit - there could be *curved* lines which give a different limiting value.

2.3.4 Properties of Limits

Lemma 50. Let $\Omega \subset \mathbb{R}^n$, $\vec{f}, \vec{g} : \Omega \rightarrow \mathbb{R}^m$, $\vec{a} \in \Omega$. For each of the following equations, if the limits on the right hand side (RHS) exist, then the limit on the left hand side (LHS) exists, and the equation holds.

(i)

$$\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) + \vec{g}(\vec{x}) = \lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) + \lim_{\vec{x} \rightarrow \vec{a}} \vec{g}(\vec{x}).$$

(ii)

$$\lim_{\vec{x} \rightarrow \vec{a}} \lambda \vec{f}(\vec{x}) = \lambda \lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}), \quad (\text{for } \lambda \in \mathbb{R}).$$

(iii)

$$\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) \vec{g}(\vec{x}) = \lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) \cdot \lim_{\vec{x} \rightarrow \vec{a}} \vec{g}(\vec{x}).$$

(iv)

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\vec{f}(\vec{x})}{\vec{g}(\vec{x})} = \frac{\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x})}{\lim_{\vec{x} \rightarrow \vec{a}} \vec{g}(\vec{x})}, \quad \text{if } \lim_{\vec{x} \rightarrow \vec{a}} \vec{g}(\vec{x}) \neq 0.$$

(v)

$$\lim_{\vec{x} \rightarrow \vec{a}} (\vec{f}(\vec{x}))^n = \left(\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) \right)^n, \quad \forall n \in \mathbb{Z}_{>0}.$$

(vi)

$$\lim_{\vec{x} \rightarrow \vec{a}} (\vec{f}(\vec{x}))^{\frac{1}{n}} = \left(\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) \right)^{\frac{1}{n}} \quad \forall n \in \mathbb{Z}_{>0}, \text{ if } \vec{f}(\vec{x}) \geq 0 \text{ near } \vec{a}.$$

Theorem 51 (Squeeze theorem). Let $f, g, h : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. If

$$g(\vec{x}) \leq f(\vec{x}) \leq h(\vec{x}),$$

for $\vec{x} \in \Omega$ near $\vec{a} \in \overline{\Omega}$, and

$$\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) = \lim_{\vec{x} \rightarrow \vec{a}} h(\vec{x}) = L,$$

for some $L \in \mathbb{R}$, then $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = L$.

Remark. We say that a statement $P(x)$ is true for $x \in \Omega$ near $a \in \overline{\Omega}$ if: $\exists \delta > 0$ such that, if $x \in \Omega$ with $x \in B_\delta(a) \setminus \{a\}$, then $P(x)$ is true.

Proof of the Squeeze theorem. Fix $\epsilon > 0$. Then there exists $\delta_1 > 0$ such that

$$0 < \|\vec{x} - \vec{a}\| < \delta_1, \quad \vec{x} \in \Omega \implies h(\vec{x}) - L < \epsilon.$$

Similarly, there exists $\delta_2 > 0$ such that

$$0 < \|\vec{x} - \vec{a}\| < \delta_2, \quad \vec{x} \in \Omega \implies L - g(\vec{x}) < \epsilon.$$

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Let $\delta_3 > 0$ be sufficiently small so that

$$0 < \|\vec{x} - \vec{a}\| < \delta_3, \quad \vec{x} \in \Omega \implies g(\vec{x}) \leq f(\vec{x}) \leq h(\vec{x}).$$

Setting $\delta = \min\{\delta_1, \delta_2, \delta_3\} > 0$, we see that, if $0 < \|\vec{x} - \vec{a}\| < \delta$ and $\vec{x} \in \Omega$, then

$$\begin{cases} f(\vec{x}) - L \leq h(\vec{x}) - L < \epsilon, \\ L - f(\vec{x}) \leq L - g(\vec{x}) < \epsilon, \end{cases} \implies |f(\vec{x}) - L| < \epsilon. \quad \square$$

We have the following Corollary to the Squeeze theorem

Corollary 1. *Let $f, g : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. If*

$$|f(\vec{x})| \leq g(\vec{x}),$$

for $\vec{x} \in \Omega$ near $\vec{a} \in \overline{\Omega}$, and $\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) = 0$, then $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = 0$.

Example 52. Consider the function $f(x, y) = x \cos\left(\frac{1}{x^2 + y^2}\right)$. Since

$$|f(x, y)| = |x| \underbrace{\left| \cos\left(\frac{1}{x^2 + y^2}\right) \right|}_{\leq 1} \leq |x|,$$

and $\lim_{(x,y) \rightarrow (0,0)} |x| = 0$, by the Squeeze theorem, $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.

Example 53. Consider the function $g(x, y) = \frac{(x-1)^2 \log x}{(x-1)^2 + y^2}$. Since

$$|g(x, y)| = |\log x| \underbrace{\left| \frac{(x-1)^2}{(x-1)^2 + y^2} \right|}_{\leq 1} \leq |\log x|,$$

and $\lim_{(x,y) \rightarrow (1,0)} |\log x| = |\log 1| = 0$, by the Squeeze theorem, $\lim_{(x,y) \rightarrow (1,0)} g(x, y) = 0$.

2.3.5 Finding Limits in Polar Coordinates

Note that in polar coordinates, $(x, y) \rightarrow (0, 0)$ is equivalent to $r \downarrow 0$.

Example 54.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2} = \lim_{r \downarrow 0} \frac{r^3(\sin^3 \theta + \cos^3 \theta)}{r^2} = \lim_{r \downarrow 0} r(\sin^3 \theta + \cos^3 \theta) = 0,$$

where the final equality comes from the bound

$$|\sin^3 \theta + \cos^3 \theta| \leq |\sin^3 \theta| + |\cos^3 \theta| \leq 2,$$

and the Squeeze theorem.

Example 55.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + xy}{2(x^2 + y^2)} = \lim_{r \downarrow 0} \frac{r^2 \cos^2 \theta + r^2 \cos \theta \sin \theta}{2r^2} = \lim_{r \downarrow 0} \frac{\cos^2 \theta + \cos \theta \sin \theta}{2},$$

which depends on θ . Therefore, the limit DNE.

Example 56.

$$\lim_{(x,y) \rightarrow (0,0)} xy \log(x^2 + y^2) = \lim_{r \downarrow 0} 2r^2 \cos \theta \sin \theta \log r.$$

Since $|2r^2 \cos \theta \sin \theta \log r| \leq |2r^2 \log r|$, and

$$\begin{aligned} \lim_{r \downarrow 0} 2r^2 \log r &= \lim_{r \downarrow 0} \frac{2 \log r}{r^{-2}} \quad (= \frac{-\infty}{\infty}) \\ &= \lim_{r \downarrow 0} \frac{2r^{-1}}{-2r^{-3}} \\ &= \lim_{r \downarrow 0} -r^2 = 0, \end{aligned}$$

where in the second equality, we have use L'Hôpital's rule. Applying the Squeeze theorem, we conclude the limit is zero.

2.3.6 Iterated Limits

Question: when are the following limits equal?

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right), \quad \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right), \quad \lim_{(x,y) \rightarrow (0,0)} f(x, y).$$

Example 57. If $f(x, y) = \frac{x+y}{x-y}$, then

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right) &= \lim_{x \rightarrow 0} \frac{x}{x} = 1; \\ \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right) &= \lim_{y \rightarrow 0} \frac{y}{-y} = -1; \\ \lim_{(x,y) \rightarrow (0,0)} f(x, y) &\text{ DNE.} \end{aligned}$$

Example 58. If

$$f(x, y) = \begin{cases} 1 & : x = y, \\ 0 & : x \neq y, \end{cases}$$

then we have that

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right) &= 0; \\ \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right) &= 0; \\ \lim_{(x,y) \rightarrow (0,0)} f(x, y) &\text{ DNE.} \end{aligned}$$

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In particular, this example shows that just because the iterated limits exist and are equal:

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right) = \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right),$$

does not imply that $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists.

Example 59. If

$$f(x, y) = \begin{cases} x \cos y^{-1} + y \cos x^{-1} & : x, y \neq 0, \\ 0 & : \text{else,} \end{cases}$$

then

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right) \text{ DNE;}$$

$$\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right) \text{ DNE;}$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0.$$

This example shows that if $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists, this does not imply that the iterated limits

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right), \quad \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right),$$

must exist. However, it is true that if all three limits exists, then they must coincide.